INCH-POUND

MIL-DTL-85097/14E (OS)
24 January 2003
SUPERSEDING
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12 June 1997

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS)
ROCKET CATAPULT MARK 1 MOD 1 (LOADED ASSEMBLY)

This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 1 Mod 1 Rocket Catapult, Loaded Assembly described herein shall consist of this specification sheet and the issue of MIL-A-85097 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation.

1. SCOPE

1.1 Scope. This specification sheet covers the requirements unique to the Mk 1 Mod 1 Rocket Catapult, Loaded Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Other Government documents, drawings, and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL657428 Propellant Grain, Mk 60 Mod 0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377
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DL4904030	Igniter, Rocket Motor, Mk 189 Mod 1
DL4904294	Igniter, Rocket Motor, Mk 250 Mod 2
DL1517547	Firing Mechanism, Rocket, Mk 28 Mod 2 Assembly
DL1517580	Catapult, Aircraft Ejection Seat, Rocket, Mk 1 Mod 1, Loaded Assembly

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The loaded rocket catapult shall conform to the documents specified herein and the applicable documents listed on DL4904294, DL4904030, DL1517547, and DL1517580.

3.2 Preproduction/production.

3.2.1 Ballistic performance. The rocket catapult shall meet the requirements of Tables I and II when test fired using an ejected weight of 317 pounds (see 6.3 for definitions).

TABLE I. Static firing requirements for booster.

Performance requirements	Firing Temperatures			
	-40°F		160°F	
	Max	Min	Max	Min
Ignition delay (ms)	30	0	30	0
Dynamic response index	18	-	22	-
Separation velocity (ft/s)	-	30	-	45
Launch time (ms)	260	-	170	-

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction.

4.1.1 Sample size. Twenty rocket catapults shall be submitted for preproduction inspection and tested in accordance with Table III of MIL-A-85097.

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TABLE II. Static firing requirements for sustainer.

Performance requirements	Firing Temperatures			
	-40°F		160°F	
	Max	Min	Max	Min
Ignition delay (ms)	30	0	10	0
Maximum resultant thrust (lb)	4220	2590	5500	3355
Impulse (lb-s)	-	1140	-	1140
Action time (ms)	555	410	400	325

4.1.2 Sample inspection.

4.1.2.1 Propellant grain Mk 60 Mod 0. The propellant grain, prior to assembly into the loaded rocket catapult, shall meet the requirements of the documents listed on DL657428. The propellant grains shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

4.1.2.2 Igniter Mk 250 Mod 2. Ballistic requirements of the igniter shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

4.1.2.3 Igniter Mk 189 Mod 1. Ballistic requirements of the igniter shall be considered acceptable when the loaded rocket catapult is in compliance with 3.2.1.

4.2 Production.

4.2.1 Sample size. Samples selected for production testing shall be as specified in table III.

4.2.2 Sample inspection. The production test sample, when in the static firing arrangement, shall be considered acceptable when the loaded rocket catapults are in compliance with 3.2.1.

4.2.3 Conditioning period. Time for conditioning prior to static firing shall be not less than 4 hours.

TABLE III. Static firing test sample size.

Lot size	Sample size	Firing temperature ±5°F	
		-40°F	160°F
0 to 40	6	3	3
41 to 65	8	4	4
66 to 110	10	5	5
111 to 150	14	7	7
151 to 300	20	10	10

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Mk 1 Mod 1 Rocket Catapult is used to eject the pilot from the A-4E/L aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. That the contract packaging requirements include the following: that exterior packing of two loaded Mk 1 Mod 1 Rocket Catapults be in accordance with DL1516349; that single unit packing be in accordance with Drawing 4904288.

6.3 Definitions.

6.3.1 Booster motor.

6.3.1.1 Ignition delay. Ignition delay (ID) is defined as the elapsed time between break wire break and the first indication of catapult thrust.

6.3.1.2. Catapult dynamic response index (DRI). Referring to MIL-S-9479, DRI is representative of the maximum compression of the vertebral column of the human body. In physical terms, the DRI is calculated by mathematically describing the human body in terms of an analogous, lumped parameter mechanical model consisting of a mass, spring, and damper. The DRI is determined from equations (1) and (2).

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$$\frac{d^2\delta}{dt^2} + 2\varsigma\omega_n \frac{d\delta}{dt} + \omega_n^2\delta = \frac{d^2z}{dt^2} \qquad (1)$$

$$DRI = \frac{\omega_n^2 \delta_{max}}{g} \qquad (2)$$

δ=compression of spring

ζ=0.224 (damping ratio of the model)

ω_n=52.9 radians/sec (undamped natural frequency of the model)

Substituting given numerical values, these equations become:

$$\frac{d^2\delta}{dt^2} + 23.7\frac{d\delta}{dt} + 2798\delta = \frac{d^2z}{dt^2}$$

$$DRI = 86.9\delta_{max}$$

6.3.1.3 Separation velocity. Separation velocity is defined as the average velocity of the simulated ejected weight propelled horizontally at catapult separation measured over the interval from 6-inches-before to 6-inches-after separation.

6.3.1.4 Launch time. Launch time is defined as the time between first indication of thrust on the axial thrust curve and strip off.

6.3.2 Sustainer motor.

6.3.2.1 Maximum resultant thrust. Maximum resultant thrust is defined as the maximum vectorial sum of all components of rocket motor thrust in any one instant.

6.3.2.2 Ignition delay. Ignition delay is the elapsed time from strip off to 10 percent of maximum thrust, as measured on the rising portion of the normal component rocket thrust trace.

6.3.2.3 Action time. Action time (t_A) is the time interval from the 10 percent of maximum thrust on the initial rise of the normal curve to the corresponding 10 percent of maximum thrust on the declining portion of the curve.

6.3.2.4 Impulse. The impulse is obtained by vectorially adding the longitudinal and normal impulse of the rocket motor taken over the action time.

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6.3.3 Strip off. Strip off occurs when the booster piston has fully seated in the booster tube, allowing booster gases to pass through the booster ports and ignite the secondary igniter and sustainer grain. This occurs at a point 32 inches into the stroke of the catapult.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85097/14E (OS)	2. DOCUMENT DATE (YYMMDD) 030124

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET CATAPULT MARK 1 MOD 1 (LOADED ASSEMBLY)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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